Group Project - Spring 2025

CPIS-334 Introduction to Software Project Management

Coordinator(s) Dr. Usman Ali Khan

13895

Student ID	2339691	Program	IS -IT-CS
Student Name	Osama Saud Aljabri		
Section	CS1 ,IT2		

Total Obtained Marks [ ] out of 30

SO	Max	Obtained Marks for SO
3	30	

Printed by : Dr. Usman Ali Khan Sunday, April 27 2025

Group Project - Spring 2025

CPIS-334 Introduction to Software Project Management

Coordinator(s) Dr. Usman Ali Khan

13895

Student ID	2138445	Program	IS -IT-CS
Student Name	Ali Mohamed Alamoudi		
Section	CS1 ,IT2		

Total Obtained Marks [ ] out of 30

SO	Max	Obtained Marks for SO
3	30	

Printed by : Dr. Usman Ali Khan Sunday, April 27 2025

Group Project - Spring 2025

CPIS-334 Introduction to Software Project Management

Coordinator(s) Dr. Usman Ali Khan

13895

Student ID	2236322	Program	IS -IT-CS
Student Name	Muath Ali Al-Sulami		
Section	CS1 ,IT2		

Total Obtained Marks [ ] out of 30

SO	Max	Obtained Marks for SO
3	30	

Printed by : Dr. Usman Ali Khan Sunday, April 27 2025

CPIS-334
(Introduction to Software Project Management)
Project Report Rubric
Spring-2025 Term-2

Project Grades: 30

Student ID: 2339691 Section: CS1

NEW SO	CLO Description	Activities Evaluated	Marks Assigned	Marks Obtained	Total
3	Administer project Time with appropriate techniques and tools	Setting up duration (time)	4		/12
		Setting up milestones	4		
		Configuring sequence of execution (relationships)	4		
3	Articulate the importance of good project communications on projects both internal to the team, and external to other project stakeholders.	Team management	2		/15
		Work distribution	2		
		Report writing skills	4		
		Project Presentation skills	7		
3	Effectively manage and plan key human resource functions within organization. Examine current issues, trends, practices, and processes in HRM.	Project Organizational Chart	1		/3
		---------- Role and responsibility assignments	1		
		---------- Staffing Management Plan	1		

CPIS-334
(Introduction to Software Project Management)
Project Report Rubric
Spring-2025 Term-2

Project Grades: 30

Student ID: 2138445 Section: CS1

NEW SO	CLO Description	Activities Evaluated	Marks Assigned	Marks Obtained	Total
3	Administer project Time with appropriate techniques and tools	Setting up duration (time)	4		/12
		Setting up milestones	4		
		Configuring sequence of execution (relationships)	4		
3	Articulate the importance of good project communications on projects both internal to the team, and external to other project stakeholders.	Team management	2		/15
		Work distribution	2		
		Report writing skills	4		
		Project Presentation skills	7		
3	Effectively manage and plan key human resource functions within organization. Examine current issues, trends, practices, and processes in HRM.	Project Organizational Chart	1		/3
		---------- Role and responsibility assignments	1		
		---------- Staffing Management Plan	1		

CPIS-334
(Introduction to Software Project Management)
Project Report Rubric
Spring-2025 Term-2

Project Grades: 30

Student ID: 2236322 Section: CS1

NEW SO	CLO Description	Activities Evaluated	Marks Assigned	Marks Obtained	Total
3	Administer project Time with appropriate techniques and tools	Setting up duration (time)	4		/12
		Setting up milestones	4		
		Configuring sequence of execution (relationships)	4		
3	Articulate the importance of good project communications on projects both internal to the team, and external to other project stakeholders.	Team management	2		/15
		Work distribution	2		
		Report writing skills	4		
		Project Presentation skills	7		
3	Effectively manage and plan key human resource functions within organization. Examine current issues, trends, practices, and processes in HRM.	Project Organizational Chart	1		/3
		---------- Role and responsibility assignments	1		
		---------- Staffing Management Plan	1		

FACULTY OF COMPUTING
& INFORMATION TECHNOLOGY
KING ABDULAZIZ UNIVERSITY

FCIT
KAU

كلية الحاسبات
وتقنية المعلومات
جامعة الملك عبدالعزيز

King Abdulaziz University

Department of Computer Science

Faculty of Computing and Information Technology

CPIS-334

Multi-Category online store system

Name	ID
Osama Saud Aljabri	2339691
Ali Mohamed Alamoudi	2138445
Muath Ali Al-Sulami	2236322

Dr. Hamid Hasan

Table of Contents

1. Purpose of the Project……1
2. Project Goals and Objectives…….1
2.1. Project Problem……..1
2.2. Project Objectives……..2
2.3. Project Importance…….2
2.4. Project Scope……..3
3. Work Breakdown Structure (WBS)…….4
4. Network Diagram…….5
4.1. Use of Precedence Diagram Method (PDM)……6
5. Staffing Procedure and Plan (E-Commerce Management)……..8
5.1. Create and Release Job Advertisements…….8
5.2. Collect and Screen Resumes……8
5.3. Conduct Interviews and Assess Training Needs……8
5.4. Issue Job Offers……..9
5.5. Finalize Employment and Onboard……..9
6. Organizational Chart and Histogram (E-Commerce Management)…….10
7. Major Project Milestones (Demonstrated Using MS Project)…….11
8. Detailed Project Schedule (MS Project)……..12

1. Purpose of the Project

It is a multi-product online store system to provide an online shopping platform for customers who prefer to shop and buy products and services online. The online store allows customers to view products and services and choose what they need easily and quickly through a convenient user interface. The online store can be dedicated to selling certain products, such as fashion or electronic devices, and includes various categories. The online store aims to provide a comfortable and customized shopping experience for customers, increase customer access and increase sales and revenues, build brand and trust, data analysis and improve performance.

The purpose of this project is to design and implement a multi category online store system that brings together customers and merchants in a single, unified e commerce platform. By leveraging modern information technologies, the system will capture, store, update and manage all necessary product and user data product names, descriptions, images, prices, stock levels, as well as customer profiles, addresses and payment methods so that shoppers can browse and filter offerings, add items to a virtual cart and complete their purchases via secure electronic payment or cash on delivery iative addresses key pain points identified through exploratory interviews with small business owners and their clientele namely, the difficulty customers face in viewing full product catalogs and details on social media, the time and effort wasted traveling to physical stores, and merchants' need for an organized, easily accessible database of customer information ing a clear, intuitive interface that requires no initial registration for browsing, and a streamlined quick registration process for ordering, the system aims to save time and effort for both customers and merchants. It will display products with complete detail, enable self-service shopping, and support order fulfillment workflows ultimately granting businesses a competitive advantage through better customer insights and fostering stronger, longer term customer relationships.

2. Project Goals and Objectives

2.1. the project Problem

The project team conducted some exploratory interviews with small business owners and their customers, and through them a set of problems were reached, which are summarized as follows:

1. The customer faces difficulty in seeing all the items, product details, and prices on social media pages.
2. Wasting time and effort going to the store.
3. The merchant's need to organize his customers' data in a way that allows him to refer to it easily and quickly.

2.2. Objectives of the project

Members of the research team developed a website aimed at improving the buying and selling processes and displaying products. Based on the problem of the study, the project aims to:

1. Applying what was learned during the previous academic years.
2. Developing the system completely, integrated and proportionate to the continuous technological development.
3. Facilitating the shopping, selling and purchasing process for the project's target groups.
4. Saving time and effort despite the geographical distance of the store location.

2.2.3 The importance of project

Technological development has become one of the most important pillars that help in the buying and selling process easier and faster, and there are a group of parties benefiting from the project, which are:

First: The importance of the project to the customer

1. Facilitating the purchase process for any product the customer desires electronically, thus saving time, effort and money.
2. Do not impose a place and time on the customer and communicate with the employer at any time he wants.
3. The ability to reserve products that the user wishes to purchase from the store displays shared in the system.

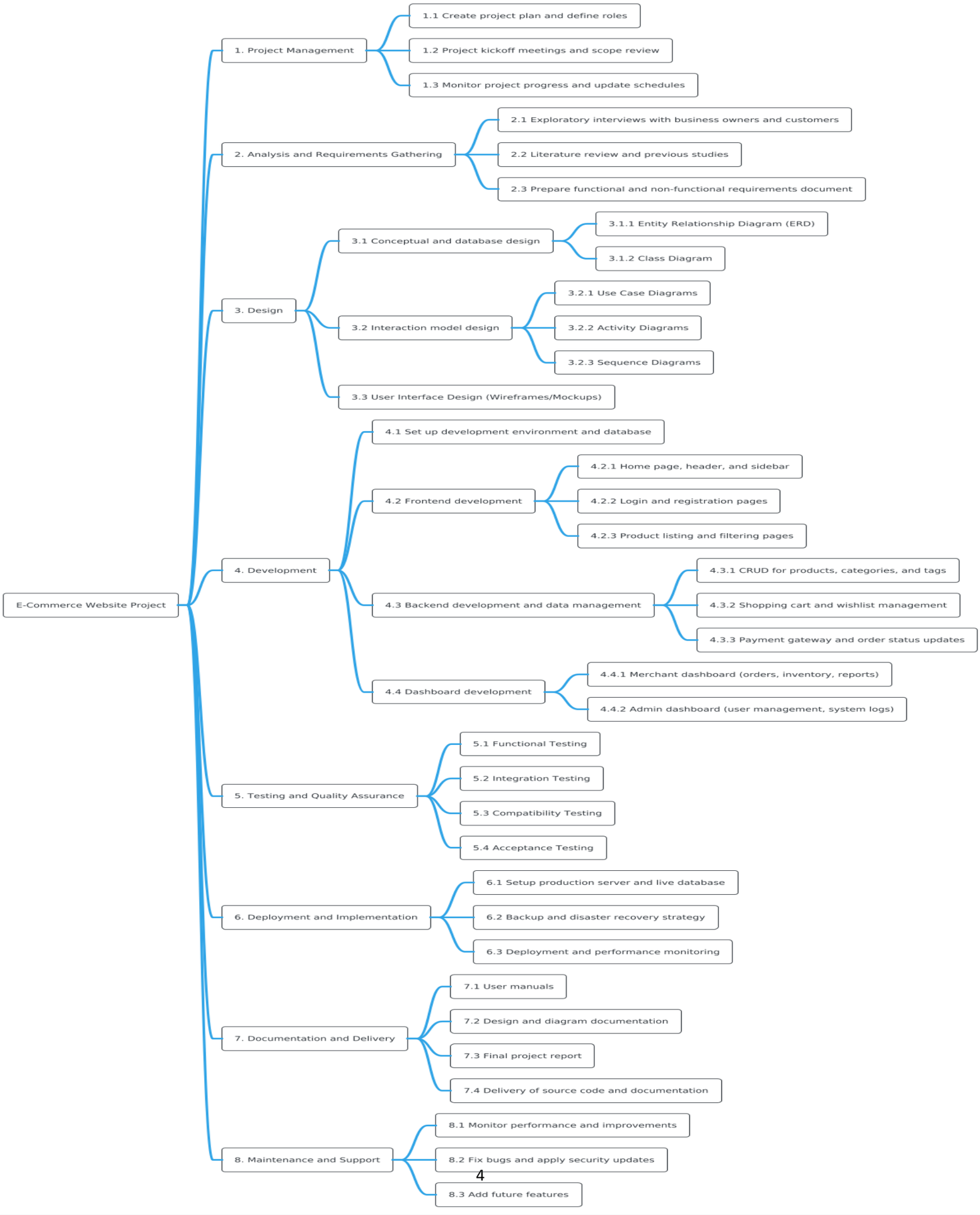
Second: The importance of the project for the merchant:

1. Show products to more users.
2. Facilitating the communication and communication process with the customer
3. Increasing profits through expanding the scope of sales.
4. Viewing the reservations made for products, which allows them to extrapolate the goods required in the future.
5. Facilitates and reduces work pressure.
6. Achieving a competitive advantage by focusing on the customer and the process of attracting him by exploiting the data available in the system.
7. Increase customer loyalty by taking their opinions into consideration.

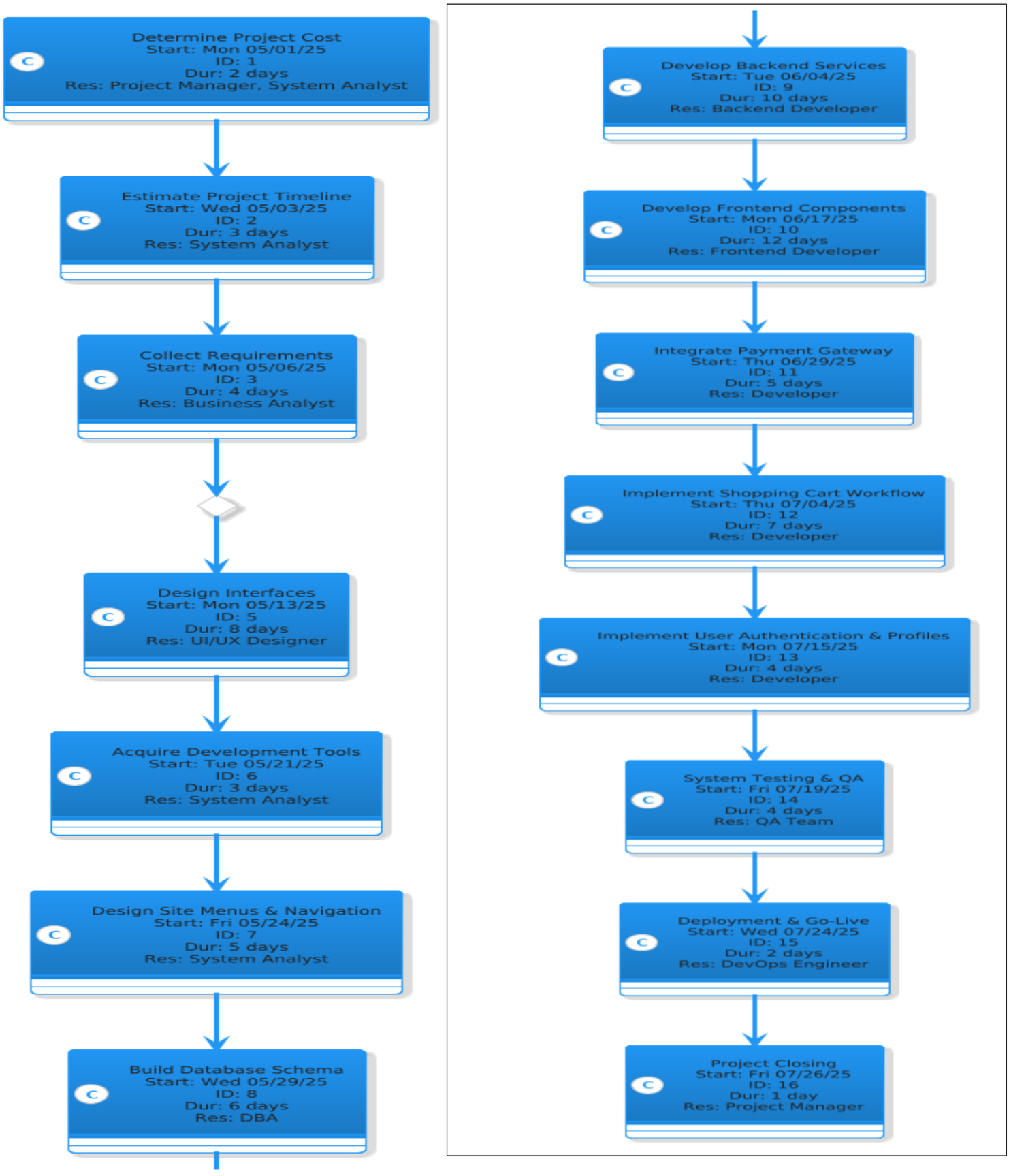
2.4. Project Scope

The e-commerce system will support both customers and merchants by providing a unified online marketplace. The project scope focuses on managing product and customer data, including functionalities to capture, store, update, and manage information such as product details (names, descriptions, images, prices, stock levels) and user profiles (personal information, addresses, payment methods), as well as shopping carts and order workflows. The system will enable customers to browse and filter products, add items to the cart, and complete purchases through integrated electronic payment gateways or the cash-on-delivery option. Merchants will be able to list and categorize products, track inventory, process orders, and access basic sales reports. Administrative features will allow system administrators to manage users, review transactions, and configure general system settings. This scope can be adjusted to meet the specific needs of any business, ranging from a single small merchant to a full multi-vendor marketplace.

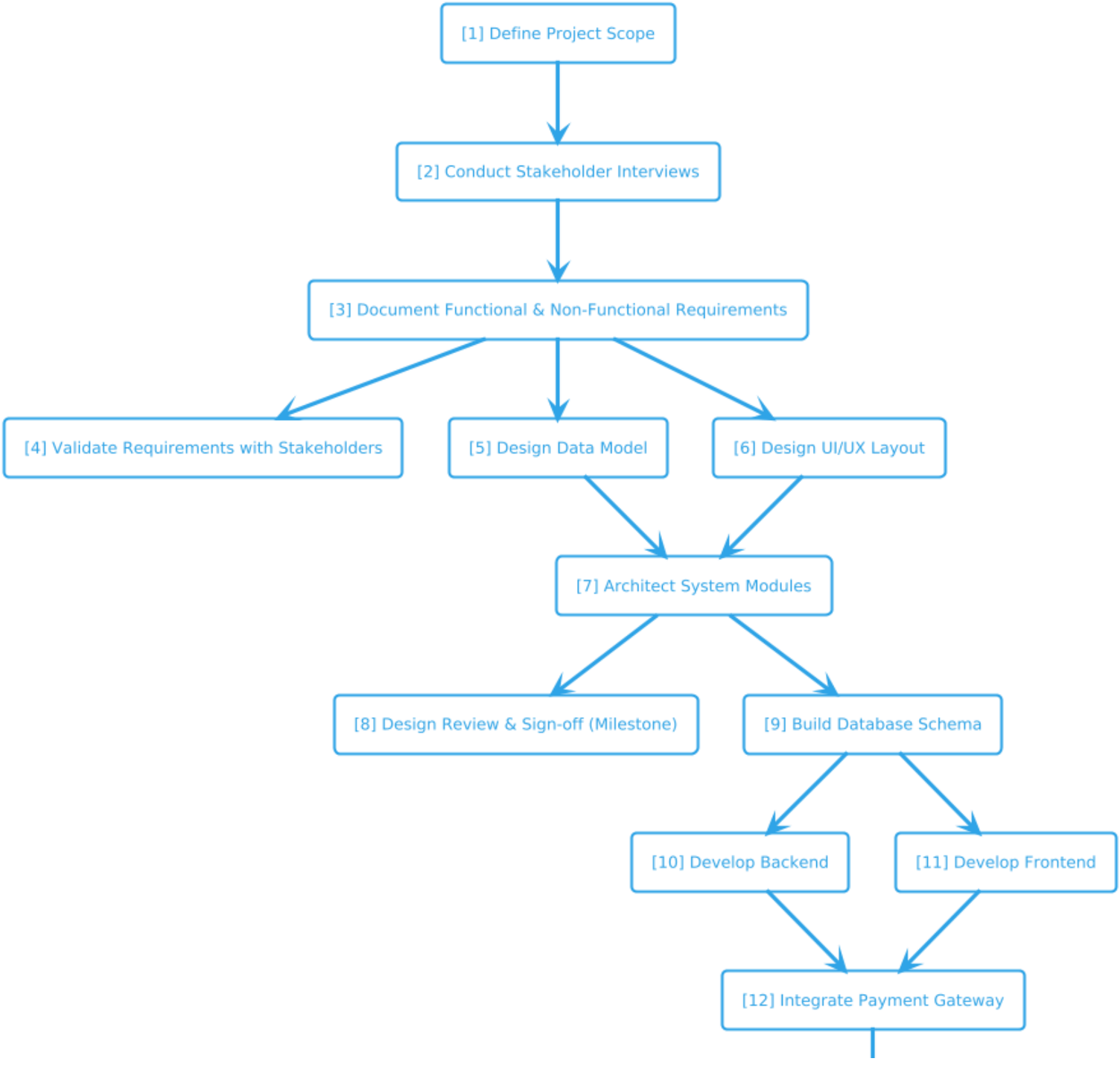
3 Work Breakdown Structure (WBS)

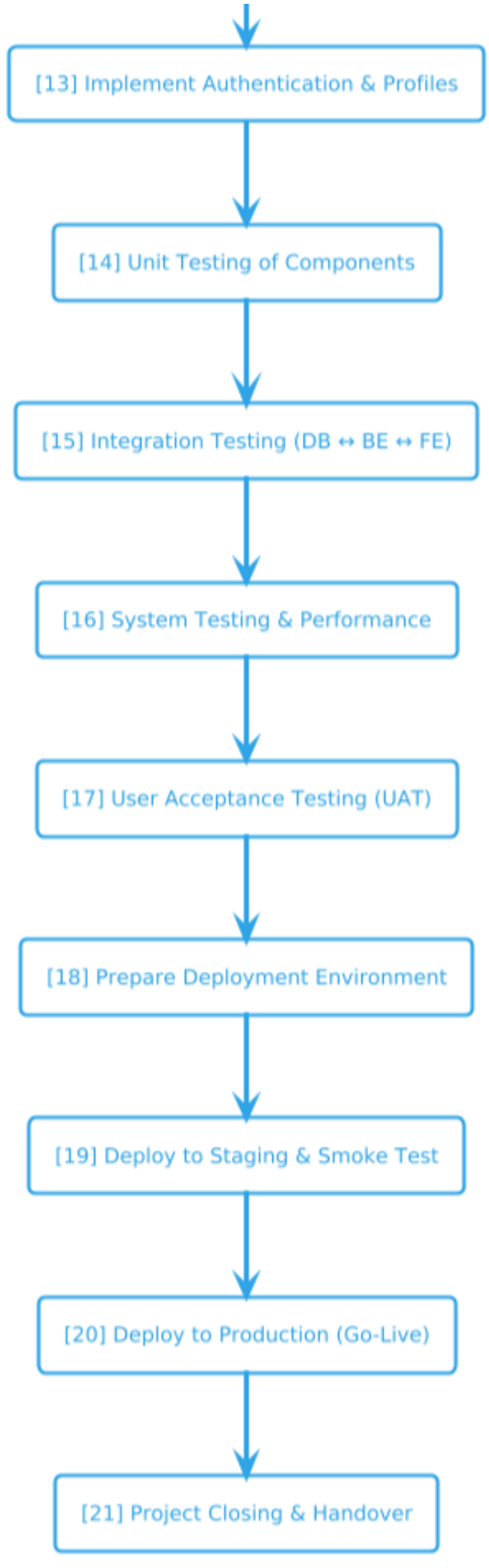

E-Commerce Website Project
1. Project Management
1.1 Create project plan and define roles
1.2 Project kickoff meetings and scope review
1.3 Monitor project progress and update schedules
2. Analysis and Requirements Gathering
2.1 Exploratory interviews with business owners and customers
2.2 Literature review and previous studies
2.3 Prepare functional and non-functional requirements document
3. Design
3.1 Conceptual and database design
3.1.1 Entity Relationship Diagram (ERD)
3.1.2 Class Diagram
3.2 Interaction model design
3.2.1 Use Case Diagrams
3.2.2 Activity Diagrams
3.2.3 Sequence Diagrams
3.3 User Interface Design (Wireframes/Mockups)
4. Development
4.1 Set up development environment and database
4.2 Frontend development
4.2.1 Home page, header, and sidebar
4.2.2 Login and registration pages
4.2.3 Product listing and filtering pages
4.3 Backend development and data management
4.3.1 CRUD for products, categories, and tags
4.3.2 Shopping cart and wishlist management
4.3.3 Payment gateway and order status updates
4.4 Dashboard development
4.4.1 Merchant dashboard (orders, inventory, reports)
4.4.2 Admin dashboard (user management, system logs)
5. Testing and Quality Assurance
5.1 Functional Testing
5.2 Integration Testing
5.3 Compatibility Testing
5.4 Acceptance Testing
6. Deployment and Implementation
6.1 Setup production server and live database
6.2 Backup and disaster recovery strategy
6.3 Deployment and performance monitoring
7. Documentation and Delivery
7.1 User manuals
7.2 Design and diagram documentation
7.3 Final project report
7.4 Delivery of source code and documentation
8. Maintenance and Support
8.1 Monitor performance and improvements
8.2 Fix bugs and apply security updates
8.3 Add future features

4 Network diagram



4.1 Use of Precedence Diagram Method (PDM)




[13] Implement Authentication & Profiles
[14] Unit Testing of Components
[15] Integration Testing (DB ↔ BE ↔ FE)
[16] System Testing & Performance
[17] User Acceptance Testing (UAT)
[18] Prepare Deployment Environment
[19] Deploy to Staging & Smoke Test
[20] Deploy to Production (Go-Live)
[21] Project Closing & Handover

5 Staffing Procedure and Plan (E-Commerce Management)

1. Create and Release Job Advertisements

For each key role, draft a clear ad including responsibilities, required skills, and qualifications. Post these on technical job boards and the company's careers page.

Role	*Key Qualifications*
Project Manager	• Bachelor's in IT/Business Administration • PMP or equivalent • 5+ years leading software projects • Agile experience
Business Analyst	• Bachelor's in Information Systems • 3+ years in requirements gathering • Excellent communication • UML/Use Case modeling
UI/UX Designer	• Degree in Design/CS • Strong portfolio of responsive web interfaces • Proficiency with Figma/Sketch
Frontend Developer	• BSc in CS or related • 3+ years with React.js, HTML/CSS, JavaScript • REST API integration experience
Backend Developer	• BSc in CS or related • 3+ years with PHP/MySQL or Node.js/PostgreSQL • Experience building secure CRUD APIs
QA Engineer	• Degree in CS • 2+ years in functional/integration testing • Familiarity with Selenium or similar • Bug-tracking tools
DevOps Engineer	• Degree in CS/Engineering • Experience with AWS/Azure, CI/CD pipelines, Docker/Kubernetes
Technical Writer	• Degree in English/Technical Communication • Experience producing user manuals and API docs

These roles map directly to the system's functional modules and technology stack (e.g., data management, UI/UX, APIs).

2. Collect and Screen Resumes

- **Resume Portal:** Set up an applicant database and intake form aligned with the advertised criteria.
- **Initial Screening:** HR reviews each resume against must-have items (e.g., years of experience, required technologies).
- **Shortlisting:** Tag candidates as "Qualified," "Consider," or "Reject," and schedule those in the first two categories for interviews.

3. Conduct Interviews and Assess Training Needs

- **Technical Interviews:**

    - **Developers:** Live coding or take-home assignment covering CRUD operations and API integration.
    - **Designer/Analyst:** Walkthrough of past deliverables, scenario-based questions (e.g., design a product-listing page).
- **Behavioral Interviews:** All candidates meet the Project Manager to evaluate teamwork, communication, and problem-solving.
- **Training Assessment:** For hires with minor skill gaps (e.g., unfamiliarity with project-specific tools), document required up-skilling modules.

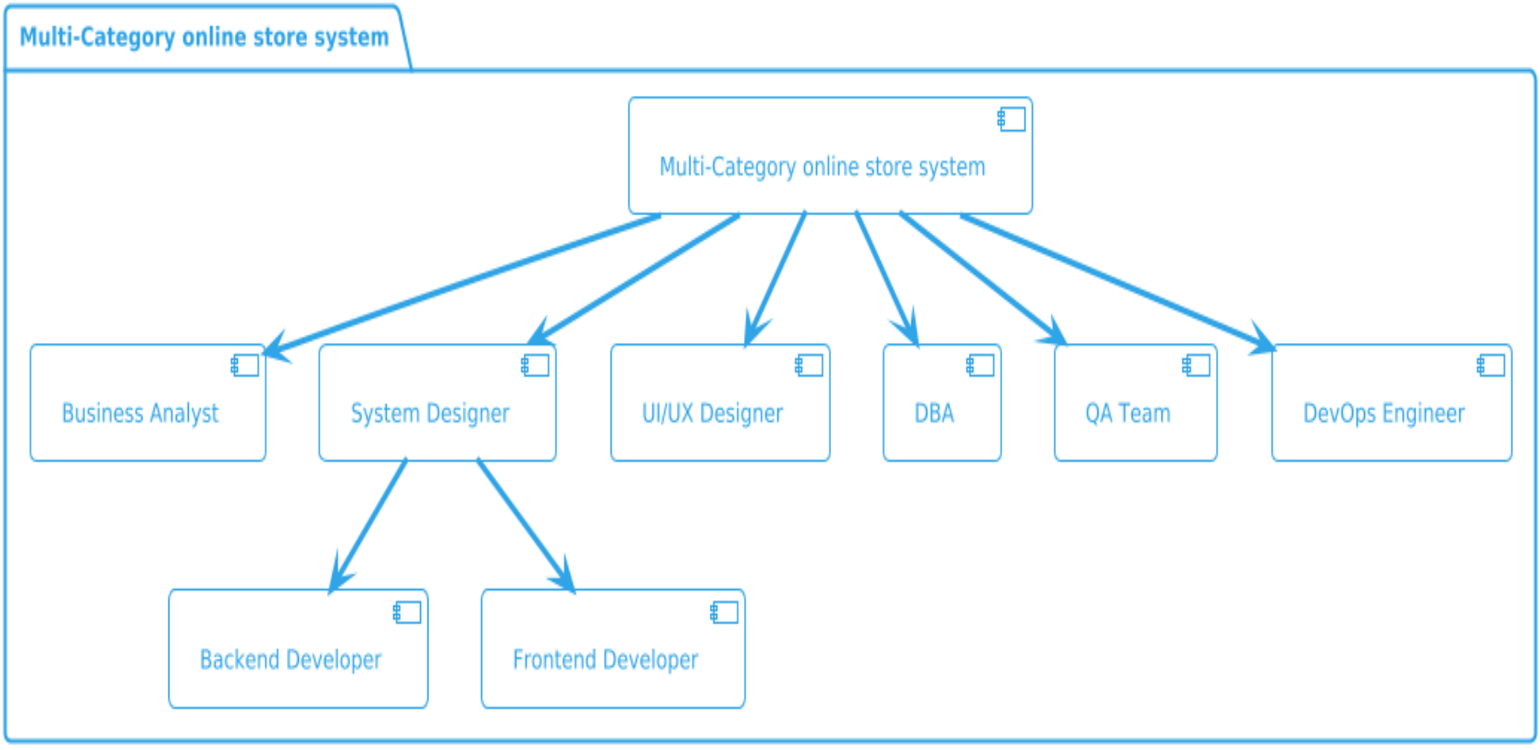
4. Issue Job Offers

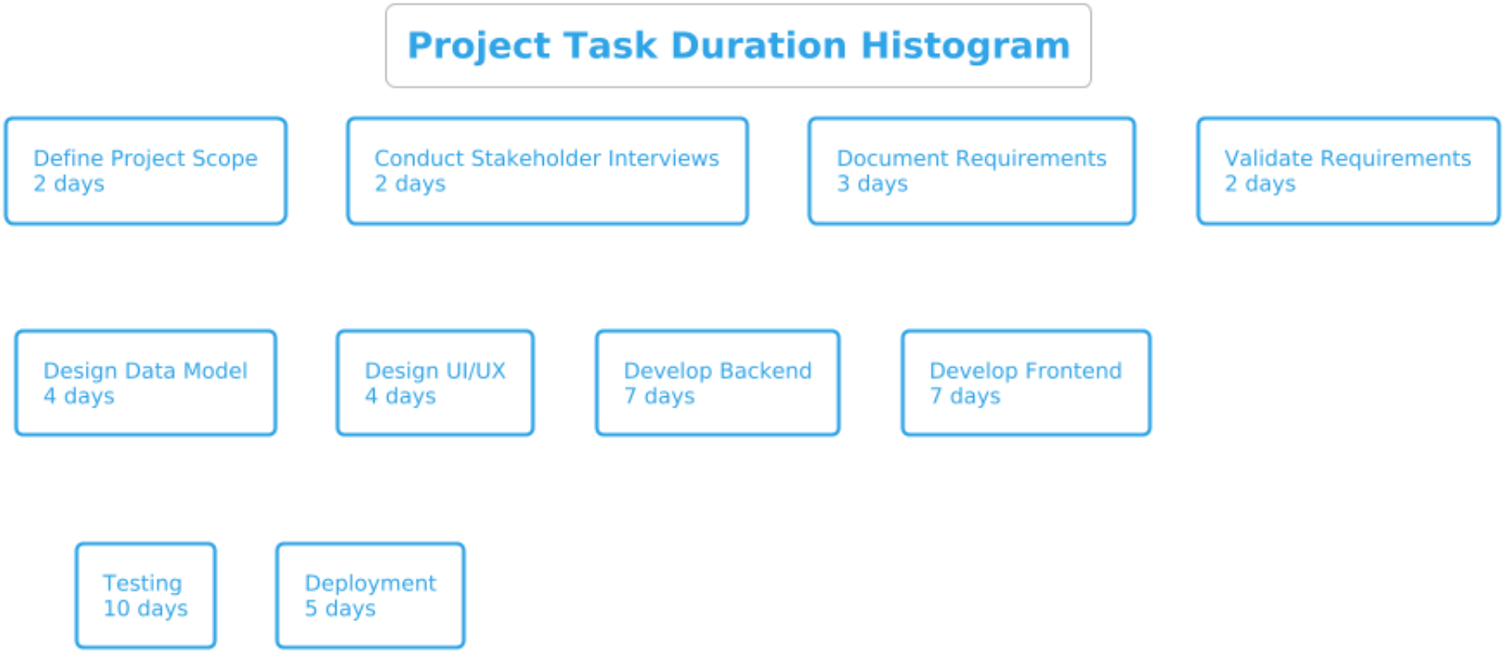
- **Offer Package Includes:**
    - **Position & Reporting Line** (e.g., "Frontend Developer reporting to Technical Lead")
    - **Salary & Compensation** (base salary, performance bonus structure)
    - **Start Date** (e.g., "June 1, 2025")
    - **Benefits** (health insurance, paid time off, training budget)
- **Approval Workflow:** Send draft offer to Finance and Legal for sign-off before extending to the candidate.

5. Finalize Employment and Onboard

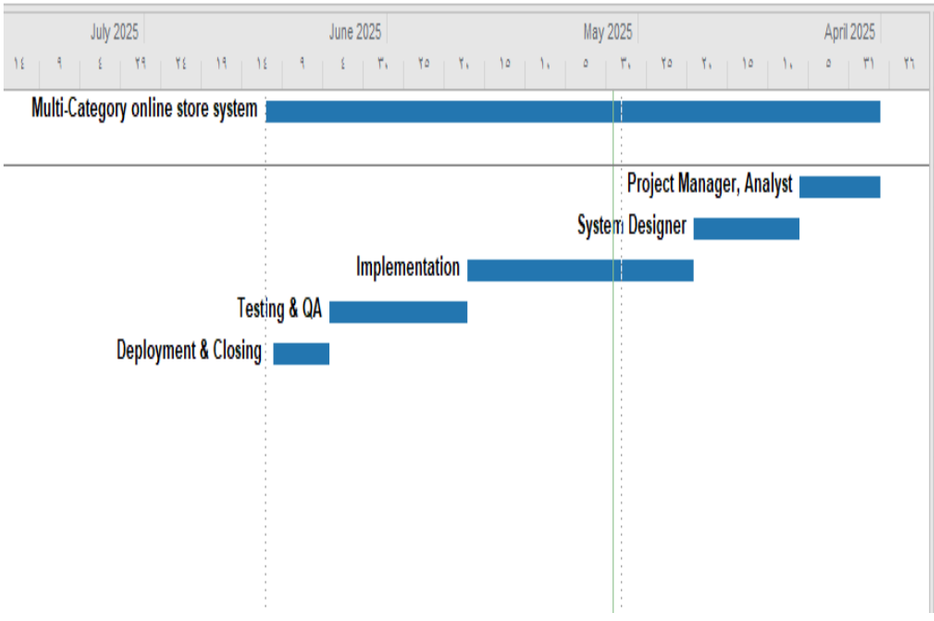
- **Employment Agreement:** Prepare contracts with standard clauses (confidentiality, IP assignment, probation period).
- **Pre-boarding:** Share company handbook, project overview, and development process docs (coding standards, environment setup).
- **First-Day Orientation:**
    - Office tour or virtual meet-and-greet with team
    - Setup of accounts (Git repo, project management tool, email)
    - Overview session on the e-commerce platform architecture and current sprint backlog

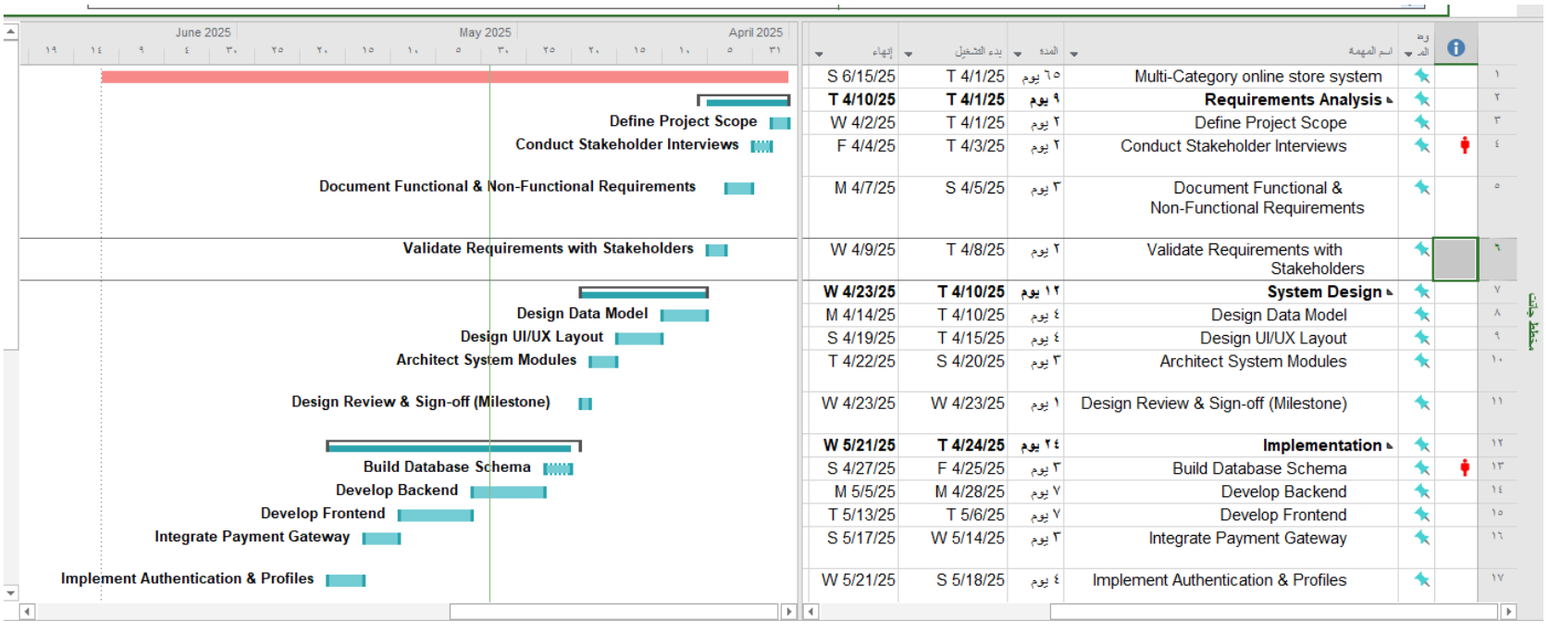
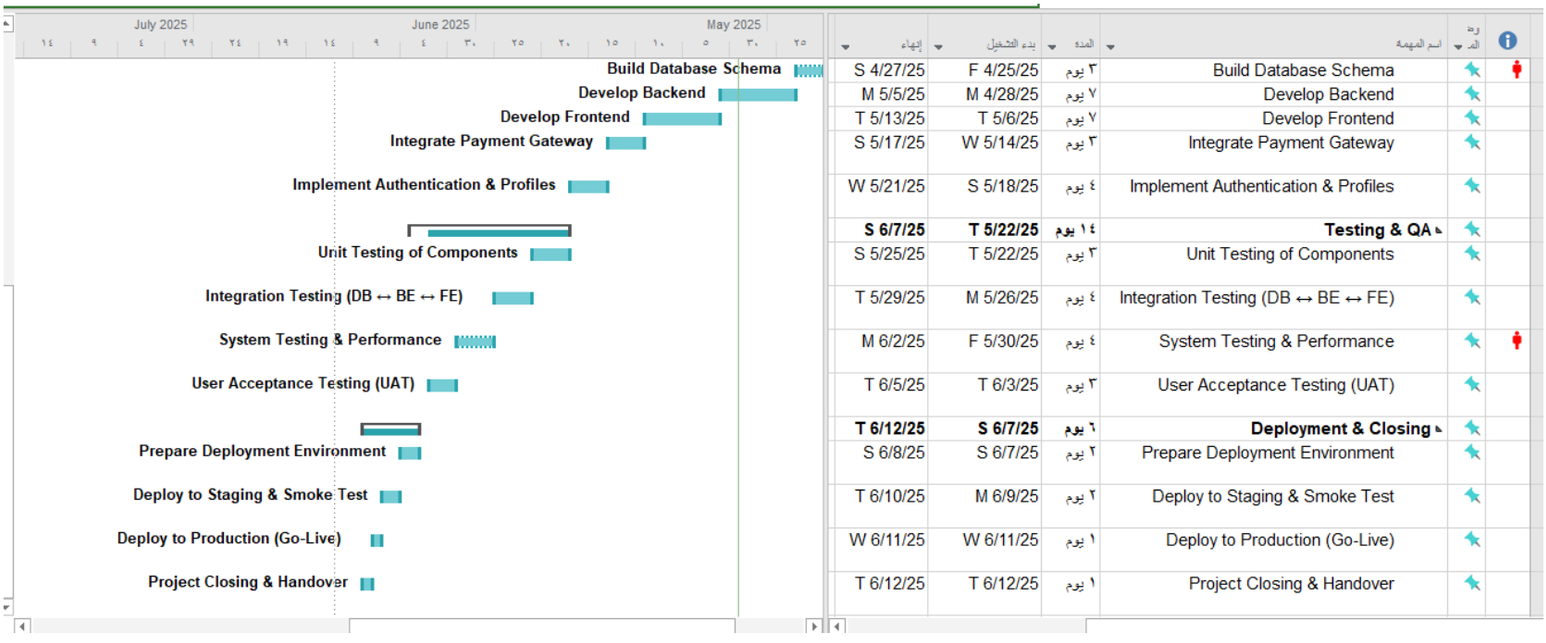
6 Organizational chart and histogram (E-Commerce Management)





7 Major Project Milestones (Demonstrate using MS Project)



وضع المهمة	اسم المهمة	المدة	بدء التشغيل	إنهاء	المهام السابقة
	Multi-Category online store system	٦٥ يوم	T 4/1/25	S 6/15/25	
	Requirements Analysis	٩ يوم	T 4/1/25	T 4/10/25	
	System Design	١٢ يوم	F 4/11/25	W 4/23/25	
	Implementation	٢٤ يوم	T 4/24/25	W 5/21/25	
	Testing & QA	١٤ يوم	T 5/22/25	S 6/7/25	
	Deployment & Closing	٦ يوم	S 6/8/25	S 6/14/25	

8 Detailed Project Schedule (MS Project)